Figure 5

Council review: oxygen placement and connected-stage exposure
Chemistry: Protect the oxygen-free Giese stage; introduce O2 only before oxidation.

Kinetics: Compare stage exposures without assuming flow acceleration or predicting yield.

Fluidics: Check inlet O2 delivery and the reverse-flow path toward Reactor 1.

Safety: Gas-line protection does not protect the liquid branch; retain the unresolved warning.

Skeptic: Do not equate numerical closure with safe operation or proven conversion.

Chief: Select a compact baseline or extend Stage 1 exposure under the conversion-focused brief.

Selected proposals: Sets 1/3: R1 = 100 min. Set 2: R1 = 250 min. All sets: R2 inlet/STP index = 181.82 min; O2 = 0.09 mL/min at STP (2.008 equiv).

Compact alternative: Stage-specific oxygen placement | Exposure comparison | Reverse-path review | Candidate selection

Figure note: Condensed from archived reports, not verbatim dialogue. Pure O2 was user-specified. Proposed conditions; backflow remained unresolved. The R2 STP index is not actual operating residence time.

Caption insert: Council review of the connected Giese addition/oxidation process. Specialist reports compare chemical order, stage exposure, equipment and gas-delivery constraints; the Skeptic and Chief retain unresolved operability issues alongside the selected screening candidate. Full reports and set-specific decisions are documented in ESI Section 7.3. Pure oxygen was specified by the researcher rather than independently discovered by the council.

Figure 6

Council review: activation and amidation as one process
Chemistry: Form the thioester first; add benzylamine between reactors without intermediate isolation.

Kinetics: Compare near-batch holds with longer activation and shorter aminolysis.

Fluidics: Recalculate Stage 2 time after amine addition; close feed concentrations and molar ratios.

Safety: Review heated pressurized operation, feed solubility and pump compatibility.

Skeptic: Flag uncertain Stage 2 conversion; no residence-time choice proves the final yield.

Chief: Select a near-batch reference or a higher-feed-throughput, asymmetric screening design.

Selected proposals: Set 1: R1/R2 = 31.25/25.00 min. Sets 2/3: R1/R2 = 35.71/14.29 min. Acid feed throughput: 0.08 versus 0.14 mmol/min.

Compact alternative: Preserve addition order | Close stoichiometry | Compare stage-time allocation | Select the first screen

Figure note: Condensed from archived reports, not verbatim dialogue. Proposed conditions, not optimized yields. Shorter Stage 2 exposure remained an experimental question; feed throughput is not product throughput.

Caption insert: Council review of telescoped thioester formation and amidation. Specialist reports connect addition order and stoichiometry to the residence time in each reactor, while comparing a near-batch reference with an asymmetric higher-feed-throughput alternative. The selected candidate is reported with its remaining chemical and operating uncertainties rather than as an established optimum (ESI Section 7.3).